

TD 6. Production of *Arthrospira platensis* in open ponds

Wording

Microalgae (including cyanobacteria) are now seen as a means of producing food for animals as well as high value-added molecules. Among them, *Arthrospira platensis* has now reached commercial deployment under the trade name Spirulina. Several bioreactors can be used to cultivate this cyanobacterium, of which open ponds offer a cheap, large-scale solution (Fig. 1). However, these bioreactors have two drawbacks. Firstly, the water tends to evaporate, which increases the salinity of the growth medium and thus causes stress to the cells. Secondly, they have to operate at a constant temperature, close to the optimum for the micro-organism, in this case 35°C. Both problems will be addressed in this case study.

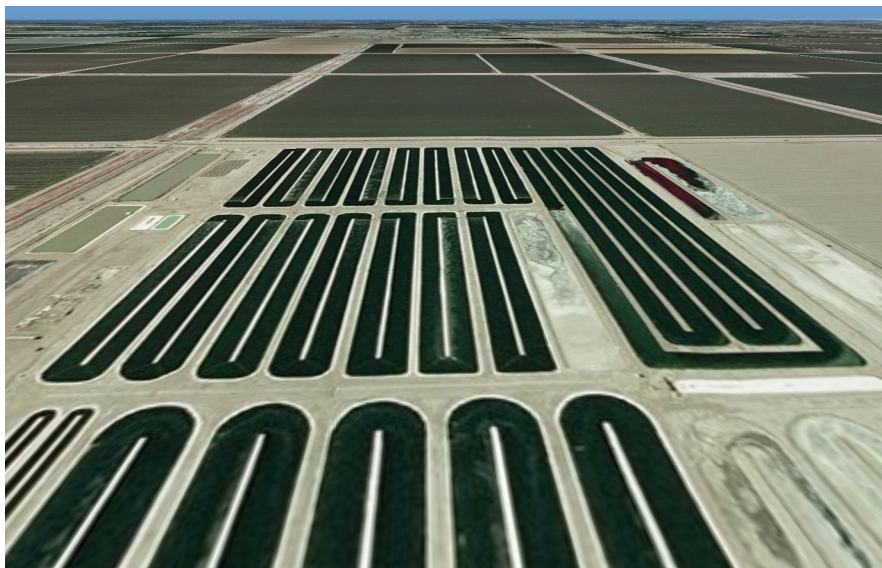


Fig. 1: Photo of raceway (credit: Duke University blogs, Nicholas School)

Assumptions

The medium is assumed to be water at a constant temperature of $T_{\text{pond}} = 35\text{ °C}$

$v_{\text{wind}} = 20\text{ km/h}$, wind speed

Thermodynamic and transport data

$T_{\text{air}} = 35\text{ °C}$, mean air temperature above the pond

$P^{\text{sat}}(35\text{ °C}) = 5626.7\text{ Pa}$, the vapour pressure of water at 35 °C

$\nu_{\text{air}} = 1.64 \cdot 10^{-5}\text{ m}^2/\text{s}$, kinematic viscosity of air at 35 °C

$D_{\text{water/air}} = 2.52 \cdot 10^{-5}\text{ m}^2/\text{s}$, mass diffusivity of water in air

$D_{\text{water/PP}} = 2.00 \cdot 10^{-13}\text{ m}^2/\text{s}$, mass diffusivity of water in polypropylene

$D_{\text{CO}_2/\text{water}} = 1.59 \cdot 10^{-9}\text{ m}^2/\text{s}$, mass diffusivity of CO₂ in water

$Y_{\infty} = 20\%$, relative humidity away from the raceway ($Y_{\infty} = C_{\infty}/C_{\text{sat}}$)

$\nu_{\text{water}} = 7.1 \cdot 10^{-7}\text{ m}^2/\text{s}$, kinematic viscosity of water at 35 °C

$\rho_{\text{water}} = 1000\text{ kg/m}^3$, masse volumique de l'eau à 35 °C

$Sc_{\text{CO}_2} = 660$, Schmidt number of CO₂

$H_{\text{CO}_2/\text{water}} = 0.031\text{ mol/L/atm}$, Henry's law constant of CO₂ in water

$\sigma = 72.1\text{ mN/m}$, surface tension between air and water

Laplace's law

$\Delta P = 2 \sigma / R$, for a bubble of radius R

Henry's law

$C_{CO_2, \text{interface}} = H_{CO_2/\text{water}} \times P_{CO_2, \text{bubble}}$ at the interface between water and bubble

Basin design and operation data

L = 50 m, length

l = 10 m, width

e = 0.30 m, depth

List of correlations

Flat plate, forced convection :

Laminar flow - averaged

$Sh_L = 0.664 Re_L^{1/2} Sc^{1/3}$ for $0.6 < Sc < 60$ and $Re < 5 \cdot 10^5$

Combined laminar and turbulent flow - averaged

$Sh_L = (0.037 Re_L^{4/5} - 871) Sc^{1/3}$ for $0.6 < Sc < 60$ and $5 \cdot 10^5 < Re < 10^7$

The dimensionless numbers are defined as:

$$Re_L = \frac{vL}{\nu}, Sc = \frac{\nu}{D}, Sh_L = \frac{k_L L}{D}, Pr = \frac{\nu}{\alpha}, Nu_L = \frac{hL}{\lambda}$$

with L the characteristic length (length, width, diameter as appropriate).

Sherwood around a bubble: $Sh = 1.35 Re^{0.5} Sc^{0.5}$

Objectives of the case study

- Assess evaporation
- Choose the bubbler for the CO₂ supply

Part 1 : Evaporation

1. Calculate the Reynolds number for the axial and cross wind configurations.
2. Calculate the Sherwood numbers for the axial and cross wind configurations.
3. Calculate the evaporation rates for the axial and cross wind configurations.

To do this, combine the equations in slides 6.6 and 6.7:

$$J_s = \frac{-D_s}{\eta} (C_i - C_\infty) \text{ and } k = \frac{-D_s}{\eta} \text{ thus } J_s = k (C_i - C_\infty) \text{ in kg/m}^2/\text{s (to be multiplied by } A_{\text{pond}})$$

4. Calculate the amount of water evaporated during cultivation (8 hours of evaporation per day for 10 days). From now on, we will consider an average evaporation rate.

In order to avoid excessive evaporation, a polypropylene sheet (200 μm thick) has been placed on top of the tank.

5. Estimate the diffusion rate of water through the polypropylene sheet. Conclude on the effectiveness of this material as a barrier.

Part 2 : CO₂ supply

To carry out photosynthesis and grow, microalgae use CO₂. In order to ensure the supply of sufficient quantities of CO₂, combustion fumes from a boiler are diluted (100 L/min at 2% CO₂ after dilution) and injected into the tank. You can choose between two bubblers, one making small bubbles (100 µm diameter), but relatively expensive, the other making larger bubbles (1 mm diameter), very affordable. We will determine which is more efficient.

Several elements on the control procedure are necessary. First, we consider that the bubbles have a speed close to their terminal velocity (0.4 m/s). The CO₂ injection is triggered when the level in the tank is below 10 mg/L and stopped when it is above 12 mg/L. The tank can therefore be considered to have a CO₂ concentration of around 11 mg/L.

6. Calculate the CO₂ concentration at the interface between the bubbles and the water, in mg/L. Use Henry's law and Young-Laplace's law to do this.

7. Calculate the resulting Reynolds number, Sherwood and transfer coefficient.

8. Calculate the CO₂ transferred by a bubble (in mg/s)

9. Calculate the number of bubbles per 1 second for each of the bubblers.

10. Calculate the total flow and conclude.